

CITYCARE HOSPITAL MANAGEMENT SYSTEM

WEEK 2: DEPARTMENT AND DOCTOR MANAGEMENT

Scenario

The CityCare Hospital Management System is a database-based application developed to manage and automate the daily operations of a hospital. For Week 2, the system focuses on maintaining information about hospital departments and doctors.

Each doctor is assigned to a specific department, and the system manages doctor details such as name, specialization, phone number, salary, and department.

The main purpose of this module is to establish a relationship between the Department and Doctor tables, apply database constraints, perform CRUD operations, and generate department-wise reports.

1. DESIGN DEPARTMENT AND DOCTOR TABLES

Department Table

The Department table stores information about the different medical departments available in CityCare Hospital.

```
SQL> CREATE TABLE Department (  
2     Department_ID NUMBER PRIMARY KEY,  
3     Department_Name VARCHAR2(100) NOT NULL UNIQUE,  
4     Location VARCHAR2(100)  
5 );
```

Table created.

Doctor Table

The Doctor table stores information about doctors and links each doctor to a department using Department_ID.

```
SQL> CREATE TABLE Doctor (  
2     Doctor_ID NUMBER PRIMARY KEY,  
3     Doctor_Name VARCHAR2(100) NOT NULL,  
4     Specialization VARCHAR2(100) NOT NULL,  
5     Phone_Number VARCHAR2(15) UNIQUE,  
6     Salary NUMBER(10,2) CHECK (Salary > 0),  
7     Department_ID NUMBER,  
8     CONSTRAINT fk_doctor_department  
9         FOREIGN KEY (Department_ID)  
10        REFERENCES Department(Department_ID)  
11 );  
  
Table created.
```

2. DEFINE RELATIONSHIPS AND CONSTRAINTS

Relationship

The relationship between Department and Doctor is **One-to-Many (1:M)**.

One department can have many doctors, while each doctor belongs to one department.

Relationship Diagram

DEPARTMENT (1) ————— (M) DOCTOR

Primary Key

Department_ID uniquely identifies each department.

Doctor_ID uniquely identifies each doctor.

Foreign Key

Department_ID in the Doctor table acts as a Foreign Key and references **Department_ID** in the Department table.

Constraints Used

PRIMARY KEY:

Uniquely identifies each record in a table.

FOREIGN KEY:

Establishes a relationship between Department and Doctor.

NOT NULL:

Ensures that required fields cannot contain NULL values.

UNIQUE:

Prevents duplicate department names and doctor phone numbers.

CHECK:

Ensures that the doctor's salary is greater than zero.

```
SQL> DESC Department;
Name                               Null?    Type
-----
DEPARTMENT_ID                     NOT NULL NUMBER
DEPARTMENT_NAME                   NOT NULL VARCHAR2(100)
LOCATION                           VARCHAR2(100)

SQL> DESC Doctor;
Name                               Null?    Type
-----
DOCTOR_ID                         NOT NULL NUMBER
DOCTOR_NAME                       NOT NULL VARCHAR2(100)
SPECIALIZATION                   NOT NULL VARCHAR2(100)
PHONE_NUMBER                      VARCHAR2(15)
SALARY                            NUMBER(10,2)
DEPARTMENT_ID                     NUMBER
```

3. INSERT VALUES

3.1 Department Table

The following department records are inserted into the Department table.

```
SQL>
SQL> INSERT INTO Doctor
  2 VALUES (103, 'Dr. Rahul Kumar', 'Neurologist', '9876543212', 110000, 2);

1 row created.

SQL>
SQL> INSERT INTO Doctor
  2 VALUES (104, 'Dr. Meena Raj', 'Orthopedic Surgeon', '9876543213', 125000, 3);

1 row created.

SQL>
SQL> INSERT INTO Doctor
  2 VALUES (105, 'Dr. Kavitha Devi', 'Pediatrician', '9876543214', 95000, 4);

1 row created.

SQL>
SQL> INSERT INTO Doctor
  2 VALUES (106, 'Dr. Sanjay Rao', 'General Physician', '9876543215', 90000, 5);

1 row created.

SQL>
SQL> INSERT INTO Doctor
  2 VALUES (107, 'Dr. Divya Anand', 'Dermatologist', '9876543216', 100000, 6);

1 row created.

SQL>
SQL> INSERT INTO Doctor
  2 VALUES (108, 'Dr. Naveen Kumar', 'General Physician', '9876543217', 92000, 5);

1 row created.

SQL>
SQL> COMMIT;|
```

```
SQL> SELECT * FROM Department;
```

```
DEPARTMENT_ID
```

```
DEPARTMENT_NAME
```

```
LOCATION
```

```
1  
Cardiology  
Block A
```

```
2  
Neurology  
Block B
```

```
DEPARTMENT_ID
```

```
DEPARTMENT_NAME
```

```
LOCATION
```

```
3  
Orthopedics  
Block C
```

```
4  
Pediatrics
```

```
DEPARTMENT_ID
```

```
DEPARTMENT_NAME
```

```
LOCATION
```

```
Block D
```

```
5  
General Medicine  
Block E
```

```
LOCATION
```

```
Block D
```

```
5  
General Medicine  
Block E
```

```
6
```

```
DEPARTMENT_ID
```

```
DEPARTMENT_NAME
```

```
LOCATION
```

```
Dermatology  
Block F
```

```
6 rows selected.
```

```
SQL> |
```

3.2 Doctor Table

The following doctor records are inserted into the Doctor table.

```
SQL> INSERT INTO Doctor
SQL> 2 VALUES (103, 'Dr. Rahul Kumar', 'Neurologist', '9876543212', 110000, 2);
1 row created.

SQL> INSERT INTO Doctor
SQL> 2 VALUES (104, 'Dr. Meena Raj', 'Orthopedic Surgeon', '9876543213', 125000, 3);
1 row created.

SQL> INSERT INTO Doctor
SQL> 2 VALUES (105, 'Dr. Kavitha Devi', 'Pediatrician', '9876543214', 95000, 4);
1 row created.

SQL> INSERT INTO Doctor
SQL> 2 VALUES (106, 'Dr. Sanjay Rao', 'General Physician', '9876543215', 90000, 5);
1 row created.

SQL> INSERT INTO Doctor
SQL> 2 VALUES (107, 'Dr. Divya Anand', 'Dermatologist', '9876543216', 100000, 6);
1 row created.

SQL> INSERT INTO Doctor
SQL> 2 VALUES (108, 'Dr. Naveen Kumar', 'General Physician', '9876543217', 92000, 5);
1 row created.

SQL>
SQL> COMMIT;|
```

```
SQL> SELECT * FROM Doctor;
```

DOCTOR_ID	DOCTOR_NAME	SPECIALIZATION	PHONE_NUMBER	SALARY	DEPARTMENT_ID
101	Dr. Arun Kumar	Cardiologist	9876543210	120000	1
102	Dr. Priya Sharma	Cardiologist	9876543211	115000	1
103					

DOCTOR_NAME		
SPECIALIZATION		
PHONE_NUMBER	SALARY	DEPARTMENT_ID
103		
Dr. Rahul Kumar		
Neurologist		
9876543212	110000	2
DOCTOR_ID		
DOCTOR_NAME		
SPECIALIZATION		
PHONE_NUMBER	SALARY	DEPARTMENT_ID
104		
Dr. Meena Raj		
Orthopedic Surgeon		
9876543213	125000	3
DOCTOR_ID		
DOCTOR_NAME		
SPECIALIZATION		
PHONE_NUMBER	SALARY	DEPARTMENT_ID
105		
Dr. Kavitha Devi		
Pediatrician		
9876543214	95000	4

DOCTOR_NAME

SPECIALIZATION

PHONE_NUMBER SALARY DEPARTMENT_ID

 106
Dr. Sanjay Rao
General Physician
9876543215 90000 5

DOCTOR_ID

DOCTOR_NAME

SPECIALIZATION

PHONE_NUMBER SALARY DEPARTMENT_ID

 107
Dr. Divya Anand
Dermatologist
9876543216 100000 6

DOCTOR_ID

DOCTOR_NAME

SPECIALIZATION

PHONE_NUMBER SALARY DEPARTMENT_ID

 108
Dr. Naveen Kumar
General Physician
9876543217 92000 5

4. CRUD OPERATIONS

4.1 INSERT RECORD

A new doctor record is inserted into the Doctor table.

8 rows selected.

```
SQL> INSERT INTO Doctor
  2  VALUES (109, 'Dr. Anjali Menon', 'Gynecologist', '9876543218', 105000, 5);
```

1 row created.

```
SQL>
SQL> COMMIT;|
```

```
SQL> SELECT * FROM Doctor
  2  WHERE Doctor_ID = 109;
```

DOCTOR_ID	DOCTOR_NAME	SPECIALIZATION	PHONE_NUMBER	SALARY	DEPARTMENT_ID
109	Dr. Anjali Menon	Gynecologist	9876543218	105000	5

4.2 UPDATE RECORD

The salary of Doctor 109 is updated from ₹105,000 to ₹110,000.

```
SQL> UPDATE Doctor
  2  SET Salary = 110000
  3  WHERE Doctor_ID = 109;
```

1 row updated.

```
SQL>
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT * FROM Doctor
  2  WHERE Doctor_ID = 109;
```

DOCTOR_ID	DOCTOR_NAME	SPECIALIZATION	PHONE_NUMBER	SALARY	DEPARTMENT_ID
109	Dr. Anjali Menon	Gynecologist	9876543218	110000	5

4.3 DELETE RECORD

The doctor record with Doctor_ID 109 is deleted from the Doctor table.

```
SQL> DELETE FROM Doctor
  2  WHERE Doctor_ID = 109;
```

1 row deleted.

```
SQL>
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT * FROM Doctor
  2  WHERE Doctor_ID = 109;
```

no rows selected

5. DEPARTMENT-WISE DOCTOR REPORTS

5.1 Display All Doctors Under Each Department

```
SQL> SELECT
  2     d.Department_Name,
  3     dr.Doctor_ID,
  4     dr.Doctor_Name,
  5     dr.Specialization
  6 FROM Department d
  7 JOIN Doctor dr
  8 ON d.Department_ID = dr.Department_ID
  9 ORDER BY d.Department_Name;
```

DEPARTMENT_NAME

DOCTOR_ID

DOCTOR_NAME

SPECIALIZATION

Cardiology

101

Dr. Arun Kumar

Cardiologist

DEPARTMENT_NAME

DOCTOR_ID

DOCTOR_NAME

SPECIALIZATION

Cardiology

102

Dr. Priya Sharma

Cardiologist

DEPARTMENT_NAME

DOCTOR_ID

DOCTOR_NAME

SPECIALIZATION

Dermatology

107

Dr. Divya Anand

Dermatologist

```

DOCTOR_ID
-----
DOCTOR_NAME
-----
SPECIALIZATION
-----
General Medicine
      108
Dr. Naveen Kumar
General Physician

DEPARTMENT_NAME
-----
DOCTOR_ID
-----
DOCTOR_NAME
-----
SPECIALIZATION
-----
General Medicine
      106
Dr. Sanjay Rao
General Physician

DEPARTMENT_NAME
-----
DOCTOR_ID
-----
DOCTOR_NAME
-----
SPECIALIZATION
-----
Neurology
      103
Dr. Rahul Kumar
Neurologist

DEPARTMENT_NAME
-----

```

```

DOCTOR_NAME
-----
SPECIALIZATION
-----
Neurology
      103
Dr. Rahul Kumar
Neurologist

DEPARTMENT_NAME
-----
DOCTOR_ID
-----
DOCTOR_NAME
-----
SPECIALIZATION
-----
Orthopedics
      104
Dr. Meena Raj
Orthopedic Surgeon

DEPARTMENT_NAME
-----
DOCTOR_ID
-----
DOCTOR_NAME
-----
SPECIALIZATION
-----
Pediatrics
      105
Dr. Kavitha Devi
Pediatrician

8 rows selected.

```

5.2 Count Number of Doctors in Each Department

```

SQL> SELECT
  2     d.Department_Name,
  3     COUNT(dr.Doctor_ID) AS Doctor_Count
  4 FROM Department d
  5 LEFT JOIN Doctor dr
  6 ON d.Department_ID = dr.Department_ID
  7 GROUP BY d.Department_Name
  8 ORDER BY d.Department_Name;|

```

```
DEPARTMENT_NAME
```

```
-----
```

```
DOCTOR_COUNT
```

```
-----
```

```
Cardiology
          2
```

```
Dermatology
          1
```

```
General Medicine
          2
```

```
DEPARTMENT_NAME
```

```
-----
```

```
DOCTOR_COUNT
```

```
-----
```

```
Neurology
          1
```

```
Orthopedics
          1
```

```
Pediatrics
          1
```

```
6 rows selected.
```

```
SQL> |
```

5.3 Find Highest-Paid Doctor in Each Department

```
SQL> SELECT
  2     d.Department_Name,
  3     MAX(dr.Salary) AS Highest_Salary
  4 FROM Department d
  5 JOIN Doctor dr
  6 ON d.Department_ID = dr.Department_ID
  7 GROUP BY d.Department_Name
  8 ORDER BY d.Department_Name;
```

```
DEPARTMENT_NAME
```

```
-----
HIGHEST_SALARY
```

```
-----
Cardiology
      120000
```

```
Dermatology
      100000
```

```
General Medicine
      92000
```

```
DEPARTMENT_NAME
```

```
-----
HIGHEST_SALARY
```

```
-----
Neurology
      110000
```

```
Orthopedics
      125000
```

```
Pediatrics
      95000
```

```
6 rows selected.
```

```
SQL> |
```

5.4 Display Departments Having More Than 1 Doctor

```

SQL> SELECT
  2     d.Department_Name,
  3     COUNT(dr.Doctor_ID) AS Doctor_Count
  4 FROM Department d
  5 JOIN Doctor dr
  6 ON d.Department_ID = dr.Department_ID
  7 GROUP BY d.Department_Name
  8 HAVING COUNT(dr.Doctor_ID) > 1;

```

DEPARTMENT_NAME

DOCTOR_COUNT

```

Cardiology
          2

General Medicine
          2

```

5.5 Find Average Doctor Salary Department-Wise

```

SQL> SELECT
  2     d.Department_Name,
  3     ROUND(AVG(dr.Salary), 2) AS Average_Salary
  4 FROM Department d
  5 JOIN Doctor dr
  6 ON d.Department_ID = dr.Department_ID
  7 GROUP BY d.Department_Name
  8 ORDER BY d.Department_Name;

```

DEPARTMENT_NAME

AVERAGE_SALARY

```

Cardiology
    117500

Dermatology
    100000

General Medicine
    91000

```

DEPARTMENT_NAME

AVERAGE_SALARY

```

Neurology
    110000

Orthopedics
    125000

Pediatrics
    95000

```

6 rows selected.

6. SUMMARY OF THE RELATIONSHIP

The Department and Doctor tables are connected through **Department_ID**.

The Department table acts as the **parent table**, while the Doctor table acts as the **child table**.

The relationship is:

Department (1) —————< Doctor (M)

This means one department can have multiple doctors, while each doctor belongs to one department.

7. CONCLUSION

The CityCare Hospital Management System Department and Doctor module was successfully designed and implemented using SQL.

A one-to-many relationship was established between the Department and Doctor tables using primary and foreign keys. Constraints such as NOT NULL, UNIQUE, and CHECK were applied to maintain data integrity.

CRUD operations including insert, update, and delete were performed successfully. Department-wise reports were also generated using SQL queries.

This module provides a structured database foundation for managing hospital departments and doctors efficiently.